Particulars laser safety during the use in STCT-01 and STCT-02

Introduction

Laser safety is an important and it is addressed in the EC 60825:1. The lasers are used in the pulsed mode where exposure time and pulse energy are key factors and also in Continuous Wave Mode (CW) for locations of the beam. The fiber coupled lasers are to be used in STCT-01 and STCT-02 systems and are rated for the use in such a system. Any other use of the Particulars lasers is not allowed.

Pulsed Lasers

Different laser wavelengths can be used in the STCT systems, ranging from 405 nm to 1064 nm. The laser diodes used differ in the Continuous Wave Power (CW) and use single mode fibre with core of 4-6 μm.

- LA-01 IR FC (1064 nm – 50 mW CW, limited by the driver to 20 mW)
- LA-01 NIR FC (980 nm – 15 mW CW)
- LA-01 R FC (660 nm – 20 mW CW)
- LA-01 G FC (520 nm – 15 mW)
- LA-01 B FC (405 nm – 15 mW)

The power of the laser diode in the CW mode is indicated in the label on each driver. The power of the laser diodes can change in time, but would remain in the same range as indicated above with all the considerations valid.



Figure 1: Laser pulse shape. The FWHM of the pulse is 350 ps.

The driver operates laser diodes with the maximum of 500 kHz repletion rate. and produces a light with a pulse of the shape shown below, typically with 350-1000 ps tunable width (FWHM). The average power of the laser is therefore calculated as

$$< P >= t_{pulse} \cdot \nu \cdot P_{CW}$$

where ν is the repletion frequency, P_{CW} power in continuous-wave mode and t_{pulse} pulse duration. Maximum laser powers at the output of the laser are therefore ranging from ~0.05 mW to maximum of ~0.3 mW.

The pulse energy W_p for this more od CW laser diode operations is calculated as

$$W_p = t_{pulse} \cdot P_{CW}$$

and the resulting W_p <50 pJ for all wavelengths.

Power measurements

The values of maximum light power for different wavelengths was measured by using Thorlabs PM100D power meter connected to Thorlabs S401C calibrated photodiode suitable for wavelengths ranging from 190 nm to 20 μm. A special mount (Figures 2 and 3) was done preventing light leak and assuring beam spot (<3 mm in diameter) fully contained in the sensitive area of the photodiode. The PM100D was calibrated and adjusted to correct wavelengths according to instructions for accurate measurements.

The measured maximum power (500 kHz, maximum pulse width achievable) for 660 nm and 1064 nm lasers were found to be very similar of around 0.04 mW, around six times lower than predicted by above calculations. The reason is very likely due to lower pulse energy achievable in pulsed mode in comparison to that calculated from power declared in the CW mode.

The measurements with BL-01 FC were also conducted and the measured power was within expected 0.1-0.2 mW.

Pulse power measurements were conducted in the system using a fully depleted 300 μm PIN diode (V_{fd}~20 V, V_{bias}=100V) and measurement of the induced current pulse in the TCT system. A calibration of the energy scale of induced charge was done using α-particles from ^{241}Am source (5.8 MeV). The maximum pulse energy of the beam was obtained to be <15 pJ for 1064 nm laser.



Figure 2: Mount of the laser FC to the Thorlabs S401C calibrated photodiode. The beam spot was fully contained on the photodiode.

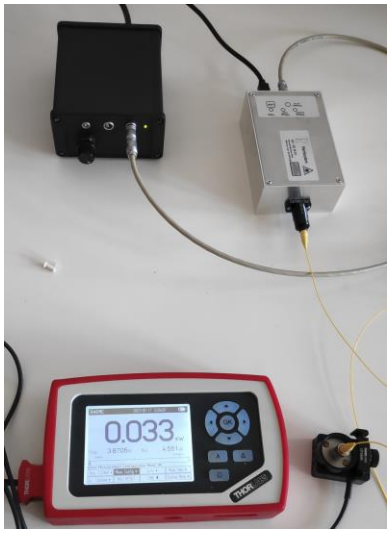


Figure 3: Measurement setup showing 1064 nm laser at maximum power and pulse width.

Optical system

The lasers should be used in the optical system of the STCT-01/02 system shown in



Figure 4: The view of the laser fiber (yellow FC on top) in the optical system. The dimension of the optical bench below is 20x20 cm². The maximum distance of the optics to the table is around 15 cm.

Figure 4. The exit from the optics system uses iris (0-1.2 cm in diameter) and has numerical aperture N_A~0.12. The minimum achievable waist of the beam coming from the optics is ~7 μm for 405 nm and ~12 μm for 1064 nm light. The system is

made in the way that direct exposure of the light to the eyes is impossible. The range of the mechanical system (box, moving tables and beam expander) is such that the closest distance to the eyes from the experimental surface is >10 cm with no possibility of direct beam exposure. Unless a special dedicated optics is used to steer the beam out of the optical system (**NOT ALLOWED**), it is impossible to have a full light pulse energy focused to better than ~1 mm^2.

Safety limits and classification of the laser

According to EC 60825:1 the Maximum permissible exposure (MPE) is given in the Figure 5. This is the defined as 10% of the power where there is 50% chance that the damage to the eye or skin can occur. With the minimum possible spot size for given optics and maximum calculated laser power the exposure values of lasers are around ~0.2-0.3 W/cm^2 (focused on retina). The measured values are around six times less (see power measurements) and by this only barely exceeding the MPE for most common exposure times shown in Figure 3. Note that for the direct exposure MPE limit is reached with the beam spot of smaller than 50 µm. Direct exposure, however, would only be possible with dismounted optics **(NEVER DO THAT)** focused directly to the eye and values exceeding MPE achieved in only few 100 microns around the beam waist (highly unlikely and also limiting the exposure time)

When laser is installed in the system the reflected beam can be treated as collimated diverging beam with >1 mm beam diameter, where MPE is $<10^{-8}$ W/cm^2, far below the maximum limits shown in Figure 3 for any exposure time.

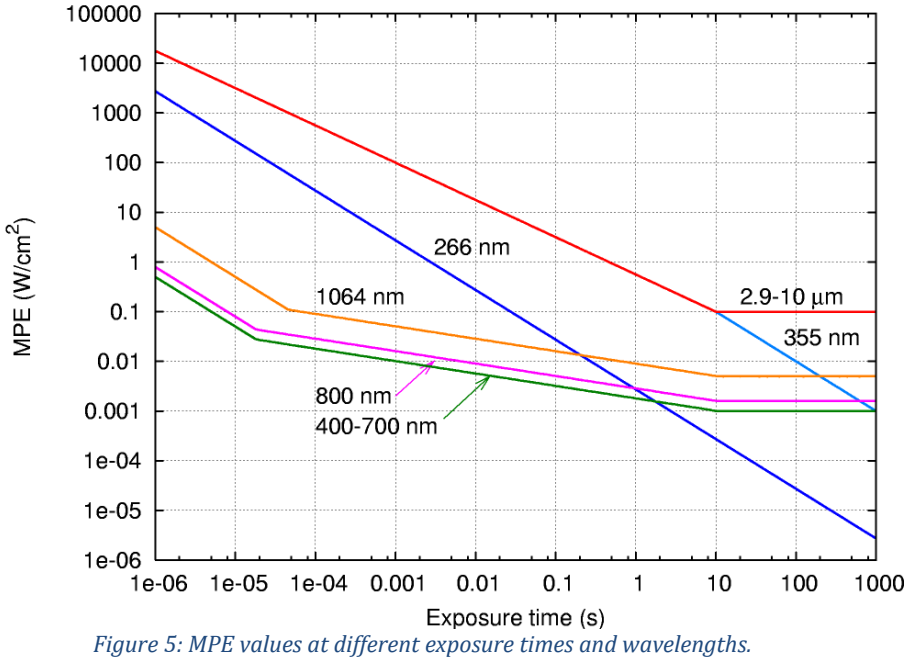


Figure 5: MPE values at different exposure times and wavelengths.

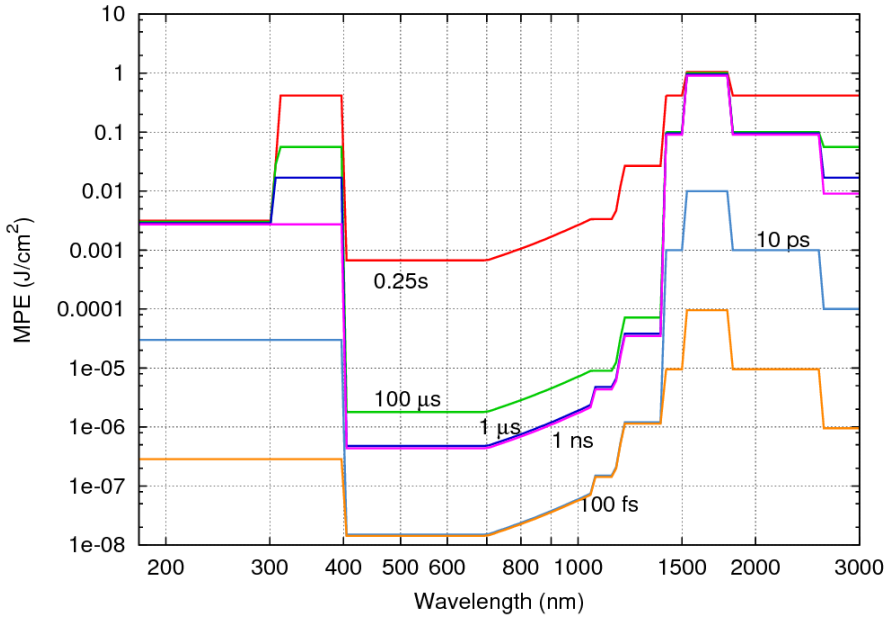


Figure 6: MPE values at different pulse durations for different pulse energies.

The MPE values per pulse are given in the Figure 6. The values for the lasers in SCT-01/02 are always below 10^{-8} J/cm^2 , hence safe.

Assuming a full pulse energy contained in the at the closest possible distance to eyes has around >1 mm^2 the accessible emission limits (AEL) were used to classify the laser.

All pulsed lasers are Class 1M lasers and are safe for all conditions when mounted in the optics.

CW lasers – beam locator

Beam locator (BL-01 FC) is laser diode used in CW mode and for the purpose of seeing visible beam spot in the region of interest on the investigated structure/device before non-visible (980 nm, 1064 nm) or less visible (405 nm) wavelengths are used.

Particulars beam locators use 635 nm red light and CW laser diode of nominally 5 mW, but the power is reduced to 0.2 mW by the circuit when in operation. A multi-mode (MM) optical fibre is used with a laser spot of 105 µm at the exit. The laser is classified as **Class 1** when installed in the system.

Safety feature of the laser power supply

The laser power supply is equipped with the lock. Please **turn the lock on** only when the laser fiber is mounted in the optical system. **Never look into the beam.**

The experimental box doesn't need the interlock switching of the laser when used with the Particulars lasers.